

FOUNDRY NETWORKS

MPLS Japan - L2 Metro
Future of MPLS L2 VPN

8 November 2002

The Power of Performance

June 2002

The Problem... Why MPLS?

- **Why do we like MPLS L2 VPN?**
- What is MPLS L2 VPN?
- When should we use MPLS L2 VPN?

2002/11/22 ©2002 Foundry Networks, Inc.

Service Provider Problems...

- Minimum ROI with Legacy Technologies
  - ATM (ah, seems like only yesterday)
  - ISDN
  - Frame Relay
  - Complexity to Install/Maintain = Higher Cost of Ownership
- Few Resources Available
  - Sluggish Economy
  - Skilled Engineers
- Making Real Profits with Networking Services
- BOTTOM Line: How much revenue does my network make?
- Time to spend wisely and be profitable. Take care of business.

FOUNDRY NETWORKS

2002/11/22 ©2002 Foundry Networks, Inc.

Solving the Problems

- Centralizing on Fewer Technologies
  - MPLS (Key NG Technology)
  - Ethernet (Champion of L2 Technologies)
  - IPv4 (Champion of L3 Technologies)
  - IPv6 (Key NG Technology)
- Combine Resources
  - Common Systems = Common Skills = Simplicity = Lower TCO
- Profitable Service Planning
  - Cost of Equipment (per port & per device)

FOUNDRY NETWORKS

2002/11/22 ©2002 Foundry Networks, Inc.

Label Switching

- Foundry switches remove the CRC from incoming Ethernet frames and replaces the 4 bytes with a Forwarding ID (FID) inside our box
  - Removing 4 bytes and adding 4 bytes for consistency of rate limiting calculations
- The FID is how the lookup engine communicates with the forwarding engine
  - The forwarding engine makes decisions based upon the FID
- Label Switching a forwarding paradigm

FOUNDRY NETWORKS

2002/11/22 ©2002 Foundry Networks, Inc.

MPLS Drivers for Service Providers

- GMPLS
  - MPLS merges legacy and future technologies
    - Shim, Native
    - Generalized packet switching
  - Multi-Protocol Lamda Switching (MPλS) O/O/O
- Forwarding Speed
  - Decisions made on labels not MAC type
  - Theoretically faster (not practically faster today)
- VPN's
- Traffic Engineering
  - QoS
  - Path Control
- Security of Provider Core

 ©2002 Foundry Networks, Inc.

Advantages of an MPLS Metro

- All Forwarding in MPLS L2 Metro is the same for broadcast, unknown unicast, known unicast, and multicast.
- Separation of Control and Data Planes
- MAC addresses not visible in the MPLS core
  - No MAC address on MPLS edge with Draft Martini VLL's
  - VPLS will have MAC learning on MPLS edge
- Superior Convergence with Fast LSP Fail-over and Switchback (sub-second)
  - Minimum loss on failure (approx 300ms*)
  - Immediate switchback (approx 100ms*)

* Confirmed by Mier Report on NetIron Metro 1500

FOUNDRY NETWORKS

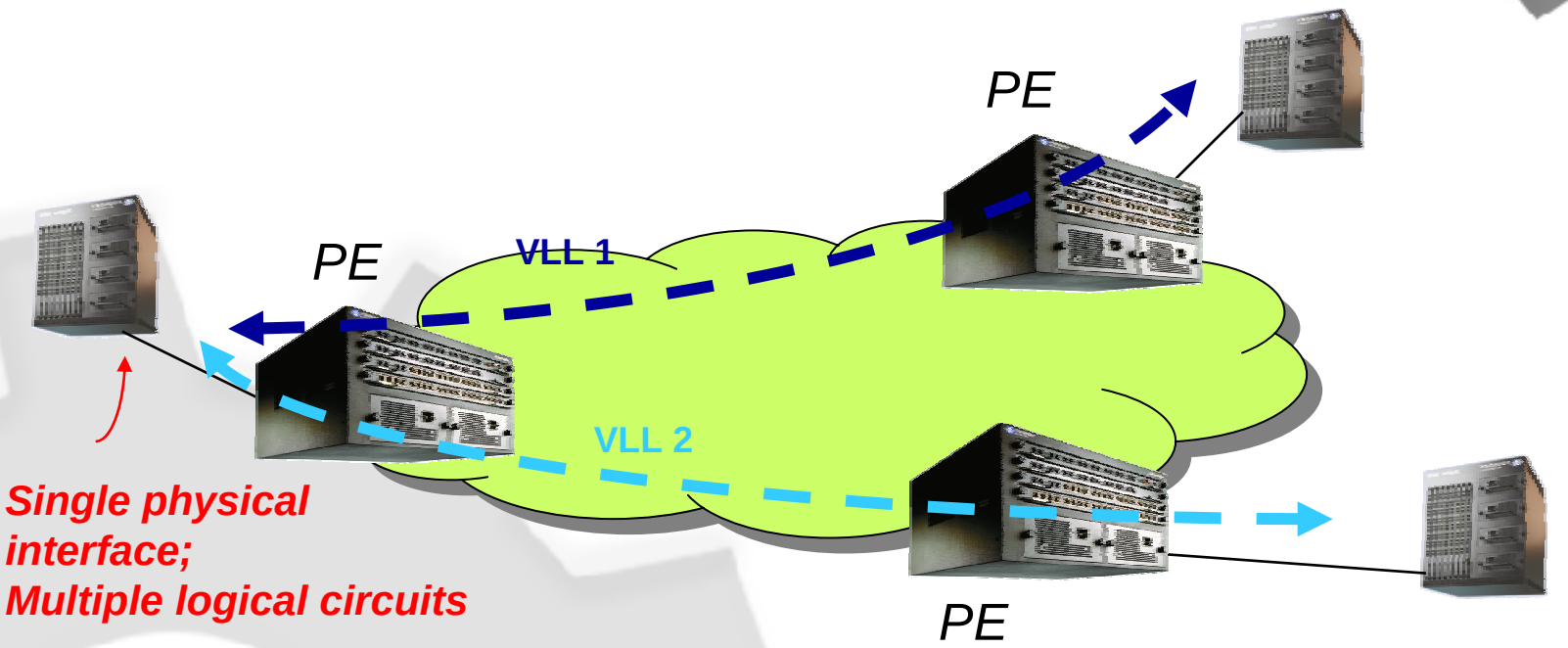
2002/11/22 ©2002 Foundry Networks, Inc.

The Problem... Why MPLS?

- Why do we like MPLS L2 VPN?
- **What is MPLS L2 VPN?**
- When should we use MPLS L2 VPN?

2002/11/22 ©2002 Foundry Networks, Inc.

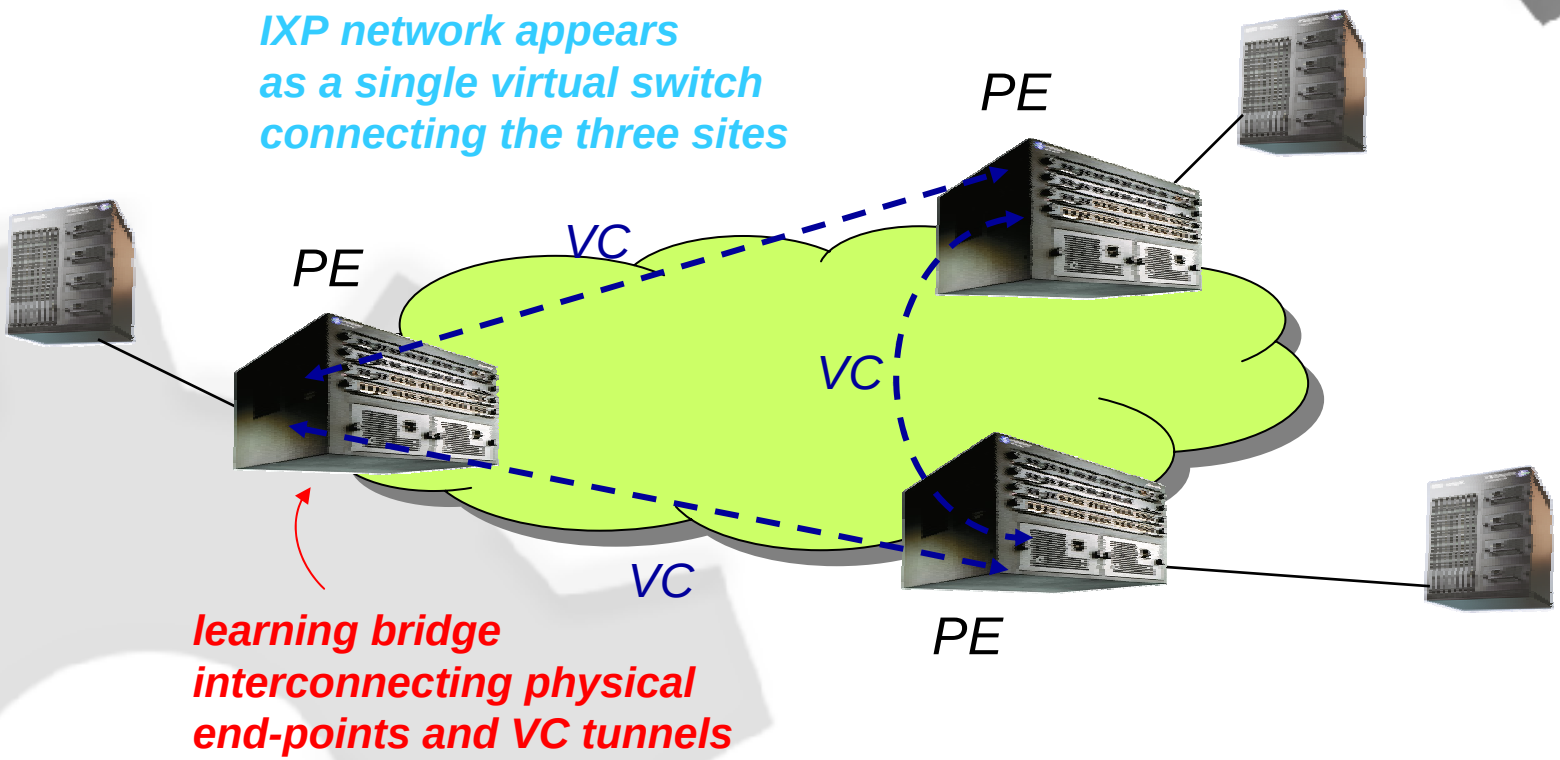
L2-VPN Solution 1: Virtual Leased Line (VLL)



IXP does not need to learn any customer MAC anywhere in its network

FOUNDRY NETWORKS
2002/11/22 ©2002 Foundry Networks, Inc.

L2-VPN Solution 2: Virtual Private LAN Service (VPLS)

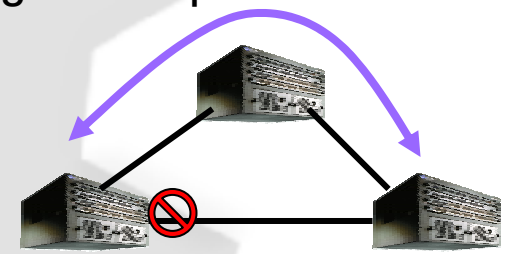


- IXP network does not need to run any L2 protocol
- Customer MAC addresses only need to be kept at edge devices – avoid large MAC tables in the core

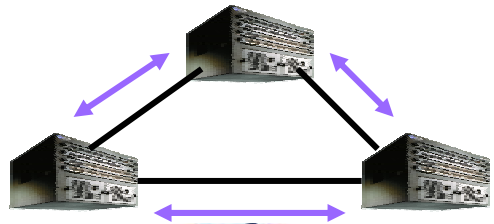
FOUNDRY NETWORKS
2002/11/22 ©2002 Foundry Networks, Inc.

Unique Benefits of L2-VPN

- Seamless upgrade from customer's point of view
- Data transport security provided by virtual circuits
  - Packets for different VPNs are distinct from each other
  - VPN customer interfaces cannot access IXP devices
- Even faster recovery from link/node failure
  - Active/standby MPLS end-to-end tunnels for rapid fail-over
  - Fail-over does not result in MAC forwarding DB aging/flushing
- Better control of traffic placement in the network
  - Shortest path forwarding edge to edge
  - Options for controlling tunnel placements



L2 topology



VPLS

2002/11/22 ©2002 Foundry Networks, Inc.

Unique Benefits of L2-VPN (cont.)

- Better traffic engineering and QoS
- VLAN-ID translation
  - Any two ends don't have to have the same VLAN-ID
  - More flexibility in VLAN management to customer
- Troubleshooting for customer connectivity problem
  - Diagnosing end-to-end connectivity
  - MPLS OAM
- Flexible intermix of VLL and VPLS solutions

FOUNDRY NETWORKS

2002/11/22 ©2002 Foundry Networks, Inc.

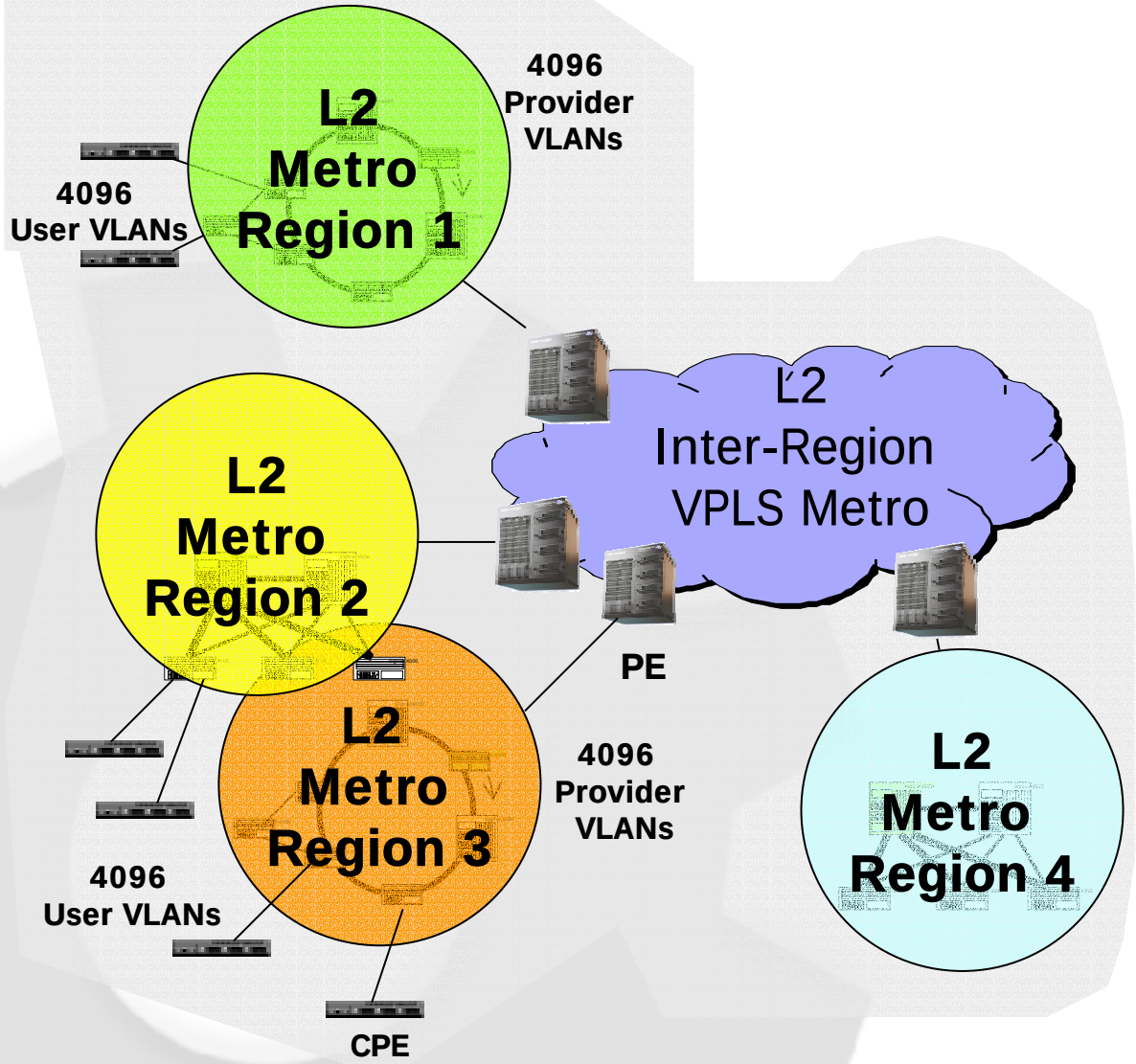
The Problem... Why MPLS?

- Why do we like MPLS L2 VPN?
- What is MPLS L2 VPN?
- **When should we use MPLS L2 VPN?**

FOUNDRY NETWORKS

2002/11/22 ©2002 Foundry Networks, Inc.

VPLS to Scale End-to-End Layer 2 Metro



- 4096 User VLANS per region with 802.1Q
- 4096 provider VLANs per region with SAV
- Solves 4096 VLAN Limit
- Regionalized MAC tables
- Efficient L2 Core with MAC scalability
- Multi-point VPN services

FOUNDRY NETWORKS

2002/11/22 ©2002 Foundry Networks, Inc.

Ethernet vs MPLS

	Ethernet	MPLS
Advanced TE	Easy to configure Topology Groups (802.1s)	Static paths, LSP's, OSPF TE, IS-IS TE, RSVP TE
High Availability	Sub-second failure with link keep alives and Rapid Spanning Tree, Metro Ring Protocol, or Virtual Switch Redundancy Protocol (MRP and VSRP are Foundry Proprietary)	Sub-second resiliency with path control
Complexity	Simple	Advanced Expert
Cost of equipment	Minimum	Maximum

FOUNDRY NETWORKS
2002/11/22 ©2002 Foundry Networks, Inc.

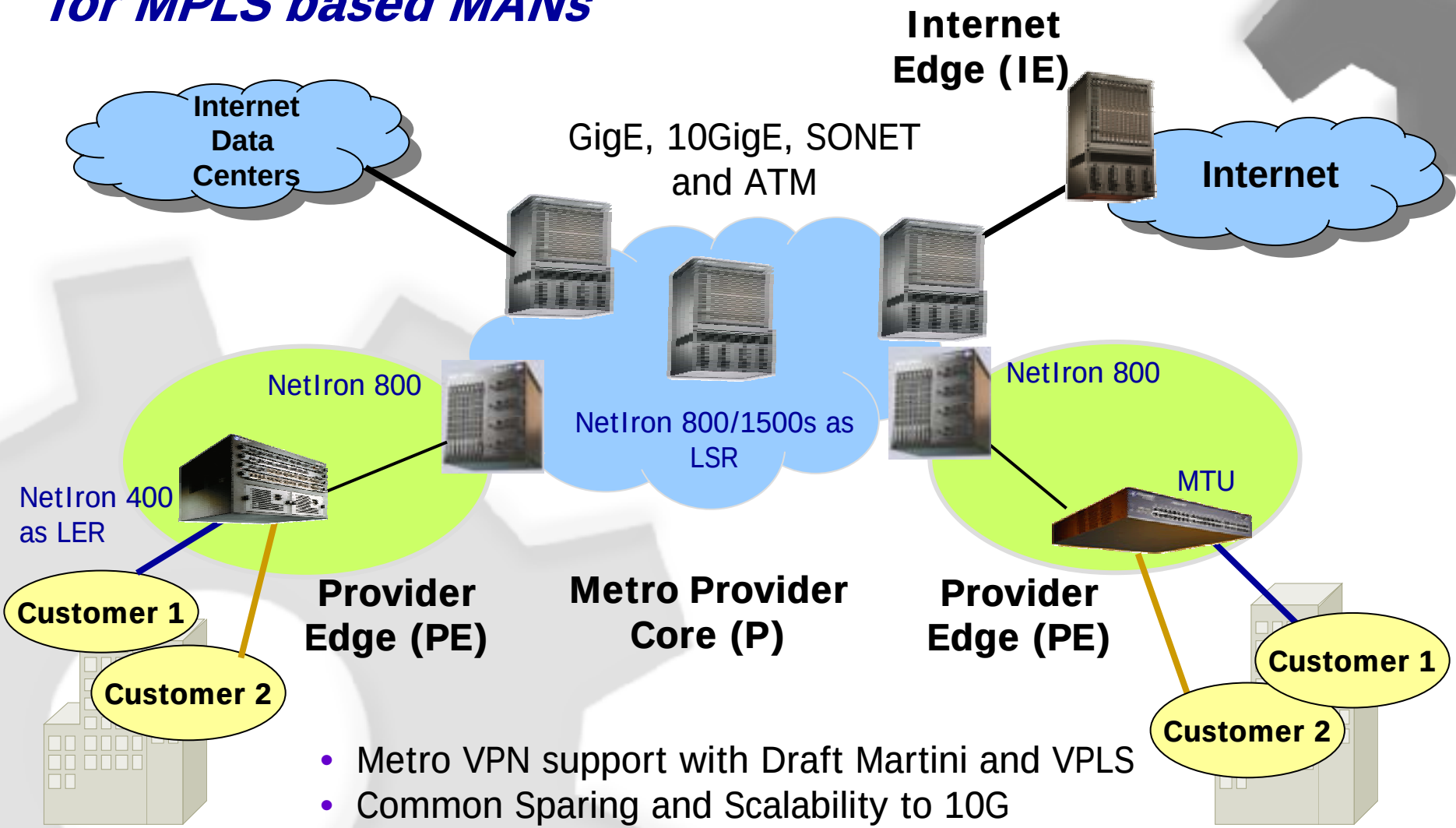
Ethernet vs MPLS and Other

	Ethernet	MPLS
Additional Encapsulation Overhead	4 or 8 bytes (VLAN and Aggregated VLAN)	4 or 8 bytes for labels + DM Encapsulation of 14 bytes (w/o control word)
Troubleshooting / Accounting / Monitoring	RFC3176 sFlow for maximum visibility of all network activities increasing my network profitability in a low cost ASIC	No equivalent yet MPLS OAM for Troubleshooting and Monitoring
Profitability	High	Low Today
Scalability	4,096 VLAN's 256 to 1M MAC's	High

2002/11/22 ©2002 Foundry Networks, Inc.

Metro Router Solution

for MPLS based MANs



- Metro VPN support with Draft Martini and VPLS
- Common Sparing and Scalability to 10G

2002/11/22 ©2002 Foundry Networks, Inc.

RFC2547 and IP VPN

- Complicated
  - VRF's, OSPF Downstream bit, Route Tagging, Address Families, SOO, RT, RD, Route Redistribution PLUS MPLS protocol suite!
  - Lots of room for mistakes (configuration)
- Costly (Complexity is directly proportional to cost)
- Many organizations will not run routing protocol with the service provider
- Lower performance (more overhead)
- More chance for bugs. ALL vendors have bugs. Protocol Interactions are difficult to troubleshoot
- NOT multi-protocol as name implies

 ©2002 Foundry Networks, Inc.

Thank You – Questions?

2002/11/22 ©2002 Foundry Networks, Inc.